

Step-1: Install and Import Required Libraries

```
# Install required libraries
!pip install transformers
!pip install torch
!pip install sentencepiece

# Import necessary libraries
from transformers import pipeline
import torch

# Verify installation
print("Transformers and Torch installed successfully!")
print("Torch version:", torch.__version__)
```

Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.25.2)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.3.0)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.22.0)
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0) (2025.10.2)
Requirement already satisfied: hf-xet<2.0.0,>=1.3.2 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0) (1.3.3)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0) (0.23.0)
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (0.24.0)
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0) (4.12.0)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0) (4.7.0)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0) (2025.11.11)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0) (1.0.7)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0) (3.10.1)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0) (0.14.0)
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0) (8.1.8)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0) (1.5.4)
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0) (13.9.0)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0) (0.0.2)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggingface-hub<2.0,>=1.3.0) (3.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggingface-hub<2.0,>=1.3.0) (2.19.1)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0->typer->huggingface-hub<2.0,>=1.3.0) (0.1.2)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.10.0+cpu)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch) (3.25.2)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)
Requirement already satisfied: sentencepiece in /usr/local/lib/python3.12/dist-packages (0.2.1)
Transformers and Torch installed successfully!
Torch version: 2.10.0+cpu

step2: Load Summarization Model

```
# Load summarization pipeline
summarizer = pipeline(
    "text-generation",
    model="facebook/bart-large-cnn"
)

print("Summarization model loaded successfully!")
```

```
Loading weights: 100% 316/316 [00:00<00:00, 561.84it/s, Materializing param=model.decoder.layers.11.self_attn_layer_norm.weight]
BartForCausalLM LOAD REPORT from: facebook/bart-large-cnn
Key | Status | |
-----+-----+
model.encoder.layers.{0...11}.final_layer_norm.bias | UNEXPECTED | |
model.encoder.layers.{0...11}.self_attn.q_proj.bias | UNEXPECTED | |
model.encoder.layers.{0...11}.fc1.bias | UNEXPECTED | |
model.encoder.layers.{0...11}.self_attn_layer_norm.weight | UNEXPECTED | |
model.encoder.layers.{0...11}.self_attn.q_proj.weight | UNEXPECTED | |
model.encoder.embed_positions.weight | UNEXPECTED | |
model.encoder.layers.{0...11}.self_attn.k_proj.weight | UNEXPECTED | |
model.encoder.layers.{0...11}.self_attn.out_proj.weight | UNEXPECTED | |
model.encoder.layers.{0...11}.self_attn.k_proj.bias | UNEXPECTED | |
model.encoder.layers.{0...11}.final_layer_norm.weight | UNEXPECTED | |
model.encoder.layers.{0...11}.self_attn_layer_norm.bias | UNEXPECTED | |
model.encoder.layers.{0...11}.fc2.bias | UNEXPECTED | |
model.encoder.layers.{0...11}.self_attn.v_proj.bias | UNEXPECTED | |
model.encoder.layers.{0...11}.fc2.weight | UNEXPECTED | |
model.encoder.layers.{0...11}.fc1.weight | UNEXPECTED | |
model.encoder.layers.{0...11}.self_attn.v_proj.weight | UNEXPECTED | |
model.encoder.layers.{0...11}.self_attn.out_proj.bias | UNEXPECTED | |
model.encoder.layer_norm_embedding.weight | UNEXPECTED | |
model.encoder.layer_norm_embedding.bias | UNEXPECTED | |

Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
Summarization model loaded successfully!
```

step3 :Prepare Input Texts

```
texts = {
    "news": """
    Artificial intelligence is transforming industries worldwide. Companies are
    using machine learning models to improve healthcare, finance, and transportation.
    Experts believe AI will create new job opportunities while also automating
    repetitive tasks.
    """,

    "technical": """
    Machine learning is a subset of artificial intelligence that enables systems
    to learn patterns from data without explicit programming. Algorithms such as
    neural networks, decision trees, and support vector machines are commonly used
    for prediction and classification tasks.
    """,

    "narrative": """
    Ravi woke up early in the morning to watch the sunrise from the hill near his
    village. As the golden rays spread across the sky, he felt a deep sense of
    peace. The birds were singing and the cool breeze made the moment unforgettable.
    """,

    "conversation": """
    Person A: Did you finish the assignment?
    Person B: Not yet, I'm still working on it.
    Person A: The deadline is tomorrow.
    Person B: I know, I'll complete it tonight.
    """
}

print("Texts loaded successfully!")

Texts loaded successfully!
```

step 4:Generate Summaries

```
for category, text in texts.items():
    summary = summarizer(text, max_length=50, min_length=20, do_sample=False)

    print("\nCATEGORY:", category.upper())
    print("Original Text:", text)
    print("Summary:", summary[0]['generated_text'])

CATEGORY: NEWS
Original Text:
Artificial intelligence is transforming industries worldwide. Companies are
using machine learning models to improve healthcare, finance, and transportation.
Experts believe AI will create new job opportunities while also automating
repetitive tasks.

Summary:
Artificial intelligence is transforming industries worldwide. Companies are
```

using machine learning models to improve healthcare, finance, and transportation. Experts believe AI will create new job opportunities while also automating repetitive tasks.
over-

CATEGORY: TECHNICAL

Original Text:

Machine learning is a subset of artificial intelligence that enables systems to learn patterns from data without explicit programming. Algorithms such as neural networks, decision trees, and support vector machines are commonly used for prediction and classification tasks.

Summary:

Machine learning is a subset of artificial intelligence that enables systems to learn patterns from data without explicit programming. Algorithms such as neural networks, decision trees, and support vector machines are commonly used for prediction and classification tasks.

CATEGORY: NARRATIVE

Original Text:

Ravi woke up early in the morning to watch the sunrise from the hill near his village. As the golden rays spread across the sky, he felt a deep sense of peace. The birds were singing and the cool breeze made the moment unforgettable.

Summary:

Ravi woke up early in the morning to watch the sunrise from the hill near his village. As the golden rays spread across the sky, he felt a deep sense of peace. The birds were singing and the cool breeze made the moment unforgettable.

CATEGORY: CONVERSATION

Original Text:

Person A: Did you finish the assignment?
Person B: Not yet, I'm still working on it.
Person A: The deadline is tomorrow.
Person B: I know, I'll complete it tonight.

Summary:

Person A: Did you finish the assignment?
Person B: Not yet, I'm still working on it.
Person A: The deadline is tomorrow.
Person B: I know, I'll complete it tonight.

step 5: Control Summary Length

```
text = texts["news"]
```

```
short_summary = summarizer(text, max_length=25, min_length=10, do_sample=False)
medium_summary = summarizer(text, max_length=50, min_length=20, do_sample=False)
long_summary = summarizer(text, max_length=80, min_length=40, do_sample=False)
```

```
print("SHORT SUMMARY:", short_summary[0]['generated_text'])
print("\nMEDIUM SUMMARY:", medium_summary[0]['generated_text'])
print("\nLONG SUMMARY:", long_summary[0]['generated_text'])
```

SHORT SUMMARY:

Artificial intelligence is transforming industries worldwide. Companies are using machine learning models to improve healthcare, finance, and transportation. Experts believe AI will create new job opportunities while also automating repetitive tasks.

MEDIUM SUMMARY:

Artificial intelligence is transforming industries worldwide. Companies are using machine learning models to improve healthcare, finance, and transportation. Experts believe AI will create new job opportunities while also automating repetitive tasks.
over-

LONG SUMMARY:

Artificial intelligence is transforming industries worldwide. Companies are using machine learning models to improve healthcare, finance, and transportation. Experts believe AI will create new job opportunities while also automating repetitive tasks.
over--- oneteteteo--o- one uner onete one-- only simply only only only- one one un--

STEP 6 — Compare Different Text Types (Analysis)

The summarization model performs differently across various text types. For news articles, the model generally produces accurate summaries because it was trained on similar datasets. Key information such as events and facts are preserved effectively. In technical texts, the model captures the main idea but may omit specialized terms or detailed explanations. Narrative texts are summarized by focusing on the central event rather than descriptive elements or emotions. Conversational texts are more challenging because they

contain informal language and short sentences, which sometimes results in overly simplified summaries. The model tends to remove filler dialogue and retain only the main topic of discussion. Overall, the summarizer performs best with structured and informative texts like news articles. However, it may struggle slightly with storytelling and conversational formats. This shows that domain-specific training can improve summarization performance.

step 7:Test with Very Long Text

```
long_text = """
Artificial intelligence has become one of the most important technological
developments of the 21st century. It is used in healthcare to assist doctors
in diagnosing diseases, in finance to detect fraud, and in transportation
to power autonomous vehicles. Researchers are continuously improving AI
algorithms to make them more efficient and reliable.

Many companies are investing heavily in AI research and development.
Technology giants are building advanced AI models capable of understanding
human language, recognizing images, and making complex decisions.

Despite its benefits, AI also raises ethical concerns. Issues such as data
privacy, algorithmic bias, and job displacement are widely discussed by
policymakers and researchers. Governments and organizations are working
towards creating regulations that ensure responsible use of AI.

In the future, AI is expected to become even more integrated into everyday
life, transforming industries and reshaping the global economy.
"""

summary = summarizer(long_text, max_length=100, min_length=40, do_sample=False)

print("LONG TEXT SUMMARY:\n", summary[0]['generated_text'])
```

LONG TEXT SUMMARY:

Artificial intelligence has become one of the most important technological developments of the 21st century. It is used in healthcare to assist doctors in diagnosing diseases, in finance to detect fraud, and in transportation to power autonomous vehicles. Researchers are continuously improving AI algorithms to make them more efficient and reliable.

Many companies are investing heavily in AI research and development. Technology giants are building advanced AI models capable of understanding human language, recognizing images, and making complex decisions.

Despite its benefits, AI also raises ethical concerns. Issues such as data privacy, algorithmic bias, and job displacement are widely discussed by policymakers and researchers. Governments and organizations are working towards creating regulations that ensure responsible use of AI.

In the future, AI is expected to become even more integrated into everyday life, transforming industries and reshaping the global economy.

STEP 8 — Lab Report Section

Objective

The objective of this experiment is to implement automatic text summarization using pretrained Transformer models from the Hugging Face Transformers library and analyze their performance across different types of text.

Dataset Description

The dataset consists of manually prepared text samples from multiple domains including news articles, technical descriptions, narrative stories, and conversational dialogue. These texts were selected to evaluate the model's ability to summarize different writing styles.

Installation and Setup

The Hugging Face Transformers library and PyTorch framework were installed using pip. The summarization pipeline was initialized using the pretrained facebook/bart-large-cnn model.

Summarization Outputs

Summaries were generated for each input text using the pipeline. The model successfully condensed long texts while preserving the key ideas.

Parameter Comparison

Different parameters such as max_length, min_length, and do_sample were adjusted to produce short, medium, and long summaries. Increasing the maximum length resulted in more detailed summaries, while shorter limits produced concise outputs.

Analysis

The model performed best on structured texts like news articles. Technical texts were summarized well but sometimes lost specific terminology. Narrative texts retained the main storyline but omitted descriptive details. Conversational texts were summarized by focusing on the main topic of discussion.

Conclusion

The experiment demonstrates that Transformer-based models are highly effective for automatic text summarization. Pretrained models allow efficient summarization without extensive training. However, the quality of summaries may vary depending on the text type and input length.